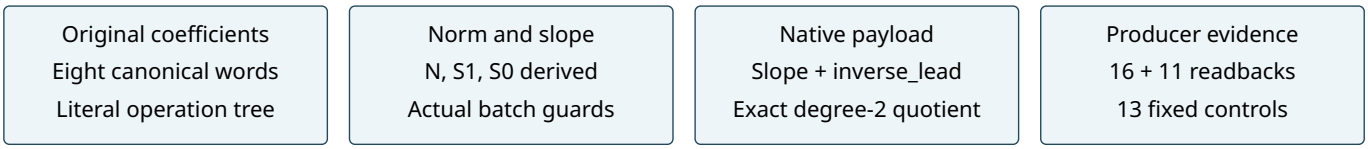
Native coefficient prelude and monic quotient payload

Two producer components connect original canonical coefficient words to the field quotient construction. They preserve the source operation order and actual norm/slope guards. Independent review is pending; original producer receipts and development failures are retained.



Coefficient prelude. An operation-tree induction proves canonicality and field decoding for the literal norm, S1 and S0 calculations from the eight original coefficient words. Explicit definitional lemmas bind the operation trees to the source definitions. The derived formulas are N=(ra-rb)^2-(qa-qb)*(ra-rb)*qb+(qa-qb)^2*rb, S1=(sb-sa)*(ra-rb)-(qa-qb)*(tb-ta), and the native S0 formula with matching ordinate orientation. Composing with the batch theorem derives canonical ni and si decoding to 1/N and 1/S1 under the actual guards. Intermediate canonicality and target formula correspondence are proved, not assumed.

Prelude checks. Nine main declarations, seven fixed controls and sixteen named readbacks passed with unchanged bindings; the false theorem was rejected. The p=17 coefficient fixture gives N=2, S1=16 and S0=7. Controls also cover the full inverse endpoint, p=3, same-support zero norm, nonzero norm with zero slope, flipped ordinate signs and excluded noncanonical input. Two retained development failures concern grouped structure syntax and an unexpanded canonicality predicate.

Literal payload. The actual slope multiplications decode to S0/N and S1/N, and the ordered inverse_lead calculation decodes to -(N/S1)^2. The decoded slope polynomial equals the field CRT slope. The source_payload_quotient endpoint constructs an exact quotient Q from the two original input residual congruences and proves degree Q=2 and monicity after scaling by the decoded inverse_lead word. It assumes neither a quotient witness nor the target slope correspondence.

Payload checks. Five main declarations, six controls and eleven named readbacks passed with unchanged bindings; the false theorem was rejected. A complete F17 word-to-monic fixture includes actual input residual witnesses. Other controls exercise p=3, nonunit values, the minus sign and the excluded zero-slope guard. One retained control attempt needed explicit polynomial-degree rewrites without changing its statement.

Domains. Prelude expression correctness uses canonical words and 0<p<2^31. The inverse and quotient endpoints require prime 2<p<2^31, actual nonzero norm and slope guards, and for the quotient degree-five f plus both input residual divisibilities. Native correspondence concerns Field.n=1 with diagnostic tables absent. No p>=5 restriction is introduced. The S0 expression represents the guarded body; it does not assert execution on bypassed branches. Polynomial validity and curve smoothness are distinct requirements.

Evidence and next composition. Source modules, check scripts, dependency lists, all attempted receipts and exact readback logs are archived separately for both components. Pinned dependency caches remain trusted; no independent binary recheck is claimed. The final scaling statement is C(decoded inverse_lead)*Q. Next compose the native polynomial multiplication, addition, exact-division, scaling and remainder models to establish the actual return values, then connect stored inputs and divisor-class interpretation. No scientific execution, timing or new cost result occurred; frozen 1,260-target scope and campaign limits remain unchanged.